BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8471

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Electrical field intensity between two oppositely charged parallel plates is:

(A) $2\sigma/\varepsilon_0$ (B) σ/ε_0

(C) $\sigma/2\varepsilon_0$ (D) ε_0/σ

Q2. When a dielectric material is inserted between the plates of a capacitor, the potential difference between the plates:

(A) Does not change (B) Increases

(C) Decreases (D) Increases then decreases

Q3. Kirchhoff's first rule is based on law of conservation of:

(A) Mass (B) Momentum

(C) Energy (D) Charge

Q4. Magnetic field strength is measured in terms of:

(A) Wbm^{-2} (B) Wb

(C) NmA^{-1} (D) Js

Q5. In CRO the output waveform of time base generator is:

(A) A ripple
(B) Square wave
(C) Sinusoidal
(D) Saw tooth

Q6. Mutual inductance of two coils does not depend on:

(A) Number of turns of the coils
(B) Area of cross-section of coils
(C) Density of material of coils
(D) Nature of the core material

Q7. If the magnetic field intensity is doubled then magnetic energy density becomes:

(A) Four times
(B) Double
(C) Half
(D) Eight times

Q8. Direct current cannot flow through:

(A) Resistor
(B) Capacitor
(C) Inductor
(D) Ammeter

Q9. In RLC series circuit, the condition for resonance is:

(A) Xc > XL
(B) Xc = XL
(C) Xc < XL
(D) XL = Z

Q10. Dimensions of strain are same as that of:

(A) Stress
(B) Pressure
(C) Young's modulus
(D) Relative permittivity

Q11. Forward resistance of the p-n junction is:

(A) Very large
(B) Of the order of kΩ
(C) A few Ohms
(D) In mega Ohms

Q12. In a transistor greater concentration of impurity is added in:

(A) Emitter
(B) Collector
(C) Both emitter and collector
(D) Base

Q13. Value of Plank's constant is:

(A) 6.34×10^{-43}Js
(B) 6.43×10^{-34}Js
(C) 6.64×10^{-19}Js
(D) 6.63×10^{-34}Js

Q14. A gamma radiation has an energy of the order of:

(A) 1 MeV
(B) 1 keV
(C) 100 eV
(D) 1 eV

Q15. Rydberg constant is given in units of:

(A) kg^{-1} (B) m^{-1}

(C) s^{-1} (D) Js

Q16. In a nuclear transmutation, radium changes into radon, the emitted particle is:

(A) A neutron (B) A proton

(C) An alpha particle (D) A beta particle

Q17. The average number of neutrons produced per fission of uranium-235 atom is:

(A) 2.5 (B) 3

(C) 2 (D) 4

Physics (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8471

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Do electrons tend to go to region of high potential or of low potential?

(ii) Define electron volt. Show that 1 eV = 1.60×10^{-19}J

(iii) Is E necessarily zero inside a charged rubber balloon if balloon is spherical? Assume that charge is distributed uniformly over the surface.

(iv) What are differences between electric force and gravitational forces?

(v) How can you use a magnetic field to separate isotopes of chemical element?

(vi) Why does a picture on a TV screen become distorted when a magnet is brought near the screen?

(vii) Define the terms magnetic flux and magnetic flux density.

(viii) Determine the magnitude of force on a charged particle in an electric and magnetic field.

(ix) Define induced current and induced emf.

(x) Show that ε and $\Delta\varphi/\Delta t$ have same units.

(xi) How the induced current can be increased?

(xii) Is it possible to change both the area of the loop and magnetic field passing through the loop and still not have an induced emf in the loop?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Describe a circuit which will give a continuously varying potential.

(ii) Explain why the terminal potential difference of a battery decrease when the current drawn from it is increased?

(iii) What is meant by tolerance? Give example.

(iv) How does doubling the frequency affect the reactance of an inductor and a capacitor?

(v) In a R-L circuit, will the current lag or lead the voltage? Illustrate your answer by a vector diagram.

(vi) Write four properties of parallel resonance circuit.

(vii) What is meant by hysteresis loss?

(viii) Define stress and strain. What are their SI units?

(ix) What are ductile and brittle substances? Give an example for them.

(x) What is the net charge on a n-type or a p-type substance?

(xi) How does the motion of an electron in a n-type substance differ from the motion of holes in a p-type substance?

(xii) What is operational amplifier?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Which has lower energy quanta? Radiowave or X-ray?

(ii) Why do not we observe Compton effect with visible light?

(iii) Define work function and give its unit.

(iv) What are the advantages of laser light over ordinary light?

(v) Define ionization potential and excitation potential.

(vi) If a nucleus has half-life of 1 year, does it mean that it will be completely decayed after 2 years? Explain it.

(vii) What do you mean by term critical mass?

(viii) Define half-life. Give its expression.

(ix) Define Hadrons and Leptons.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 05) Explain electric potential at a point due to a point charge. Derive its relation.

Q5(b). (Marks 03) A rectangular bar of iron is 2.0cm by 2.0cm in cross-section and 40cm long. Calculate its resistance if the resistivity of iron is $11\times10^{-8}\Omega m$

Q6(a). (Marks 05) Determine e/m of an electron. Derive its relation.

Q6(b). (Marks 03) Two coils are placed side by side. An emf of 0.8V is observed in one coil when current is changing at the rate of 200 As^{-1} in the other coil. What is the mutual induction of the coils?

Q7(a). (Marks 05) Describe the behaviour of an inductor in an A.C. circuit and write expression for reactance of an inductor.

Q7(b). (Marks 03) Calculate the gain of non-inverting amplifier shown in figure: [circuit diagram with 10kΩ input resistor, 40kΩ feedback resistor, input v_i].

Q8(a). (Marks 05) Describe the formation of energy bands in solids. Explain the difference of electrical behaviour of conductors, insulators and semi-conductors in terms of energy band theory.

Q8(b). (Marks 03) A sheet of lead 5.0mm thick reduces the intensity of a beam of γ-rays by a factor of 0.4. Find half value thickness of lead sheet which will reduce the intensity to half of its initial value.

Q9(a). (Marks 05) What is meant by inner shell transition and characteristics X-rays. How X-rays are produced? Write any two properties and uses of X-rays.

Q9(b). (Marks 03) An electron is accelerated through a potential difference of 50V. Calculate its de-Broglie wavelength.

GROUP – II

Physics (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8472

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. A particle of mass m and charge q is released from rest in a uniform electric field E. The K.E. attained by the particle after moving a distance 'd' is:

(A) Ed/q
(B) qE^2d
(C) qEd
(D) qE/d^2

Q2. The energy stored in the capacitor is:

(A) K.E.
(B) P.E.
(C) Electrical K.E.
(D) Electical P.E.

Q3. On increasing the length of wire specific resistance of the wire:

(A) Increases
(B) Decreases
(C) Remains unchanged
(D) First increase then decrease

Q4. An electron is moving in a circle of radius 'r' in a uniform magnetic field, suddenly the field is reduced to B/2, the radius of circle now becomes:

(A) r/2
(B) r/4
(C) 2 r
(D) 4 r

Q5. Force on current carrying conductor per unit length is given by:

(A) ILB sin θ
(B) ILB
(C) IB
(D) IB sin θ

Q6. The current flowing through a coil due to induced emf in it depends upon:

(A) Shape of the coil
(B) Resistance of the coil
(C) Area of the coil
(D) Magnetic flux

Q7. The induced emf primarily produced at the cost of:

(A) Internal energy
(B) Chemical energy
(C) Electrical energy
(D) Mechanical energy

Q8. At low frequency the current through a capacitor of A.C. circuit will be:

(A) Large
(B) Small
(C) Zero
(D) Infinite

Q9. The inductance and capacitance behave a function of:

(A) Voltage
(B) Frequency
(C) Time
(D) Current

Q10. Impurity atoms are doped in semi-conductor to increase:

(A) Free electrons
(B) Holes
(C) Conductivity
(D) Resistivity

Q11. The specially designed semi-conductor diode used as indicator lamp in electronic circuit are:

(A) The switch
(B) Solar cells
(C) Photodiodes
(D) Light emitting diode

Q12. Which diode is used for detection of light?

(A) Light emitting diode
(B) Photo diode
(C) Photo voltaic cell
(D) All these

Q13. Rest mass of photon is:

(A) Zero
(B) Infinity
(C) hf/e
(D) hc/λ

Q14. Threshold wavelength for metal having work function φ_0 is λ_0. What is threshold wavelength for metal having work function 2 φ_0 is?

(A) λ/2
(B) 4 λ
(C) 2λ
(D) λ/4

Q15. Production of X-rays can be regarded as inverse of:

(A) Compton effect
(B) Photoelectric effect
(C) Annihilation of matter
(D) Pair production

Q16. The energy released per unit mass is greater in:

(A) Fission reaction
(B) Fusion reaction
(C) Chemical reaction
(D) Nuclear reaction

Q17. Energy needed to create an electron-hole pair in a solid state detector is:

(A) 2 - 3 eV
(B) 3 - 4 eV
(C) 4 - 5 eV
(D) 5 - 6 eV

Physics (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8472

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) How can you identify that which plate of a capacitor is positively charged?

(ii) Do electrons tend to go to region of high potential or of low potential?

(iii) State Gauss's law and write its mathematical expression.

(iv) Give a comparison between electric and gravitational forces.

(v) Describe the change in magnetic field inside a solenoid carrying a steady current I, if the number of turns is doubled but the length remains the same.

(vi) If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in the region is zero?

(vii) Define magnetic flux density and write its unit.

(viii) What is Lorentz force? Write its mathematical expression.

(ix) How would you position a flat loop of wire in a changing magnetic field so that there is no emf induced in the loop?

(x) Does the induced emf always act to decrease the magnetic flux through a circuit?

(xi) Write the factors upon which mutual inductance depends.

(xii) State Faraday's law of electromagnetic induction and write its mathematical expression.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Why does the resistance of a conductor rise with temperature?

(ii) Explain why the terminal potential difference of a battery decreases when the current drawn from it is increased?

(iii) Differentiate between resistance and resistivity. Also give their units.

(iv) A sinusoidal current has 'rms' value of 10A. What is the maximum or peak value?

(v) In R-L circuit, will the current lag or lead the voltage? Illustrate your answer by a vector diagram.

(vi) At what frequency will an inductor of inductance 1.0H have reactance of 500Ω.

(vii) What is meant by hysteresis loss? How it is used in the construction of a transformer?

(viii) Define modulus of elasticity. Show that unit of modulus of elasticity and stress are same.

(ix) Differentiate between curie temperature and critical temperature.

(x) Why charge carriers are not present in the depletion region?

(xi) Why a photo diode is operated in reverse biased region?

(xii) A transistor has Ic = 10mA and IB = 40mA. Calculate the current gain.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) As a solid is heated and begin to glow, why does it first appear red?

(ii) Can pair production take place in vacuum? Explain.

(iii) What is the energy of photon in a beam of infra-red radiation of wavelength 1240nm?

(iv) Is energy conserved, when an atom emits a photon of light?

(v) What is meant by CAT-Scanner?

(vi) Why are heavy nuclei unstable?

(vii) Describe a brief account of interaction of various types of radiations with matter.

(viii) What factors make a fusion reaction difficult to achieve?

(ix) What is self-quenching in working of GM-Counter?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 05) Define capacitance. Also derive a relation for capacitance of a parallel plate capacitor for air and dielectric as a medium.

Q5(b). (Marks 03) 0.75A current flows through an iron wire with a battery of 1.5V is connected across its ends. The length of the wire is 5.0m and its cross-sectional area is $2.5\times10^{-7}m^2$. Compute the resistivity of iron.

Q6(a). (Marks 05) Derive the relation of e/m of an electron.

Q6(b). (Marks 03) Two coils are placed side by side. An emf of 0.8V is observed in one coil when the current is changing at the rate of $200As^{-1}$ in the other coil. What is the mutual inductance of the coils?

Q7(a). (Marks 05) What is an operational amplifier? Derive a relation for gain of operational amplifier as inverting amplifier.

Q7(b). (Marks 03) Find the capacitance required to construct a resonance circuit of frequency 1000kHz with an inductor of 5mH.

Q8(a). (Marks 05) Define and explain fusion reaction in detail.

Q8(b). (Marks 03) A 1.0m long copper wire is subjected to stretching force and its length increases by 20cm. Calculate the tensile strain and the percent elongation which the wire undergoes.

Q9(a). (Marks 05) What is de-Broglie hypothesis? How Davisson and Germer verify it? Explain.

Q9(b). (Marks 03) The wavelength of K X-ray from copper is $1.377\times10^{-10}m$. What is the energy difference between the two levels from which this transition results?